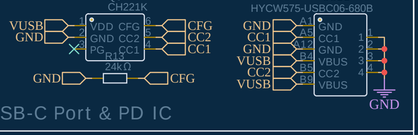
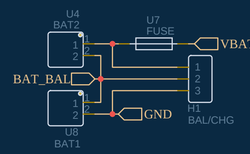
USB-C Port & PD IC

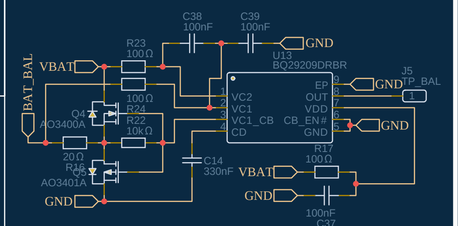


Battery Input

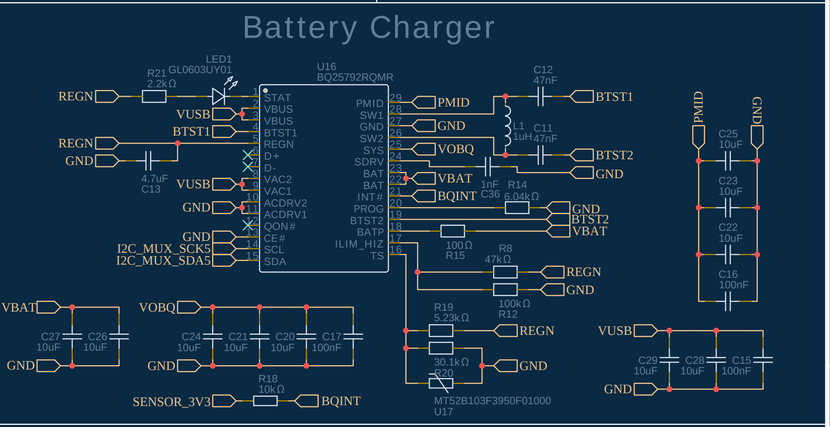
2s (2x 3.7V 5Ah Li-Po)



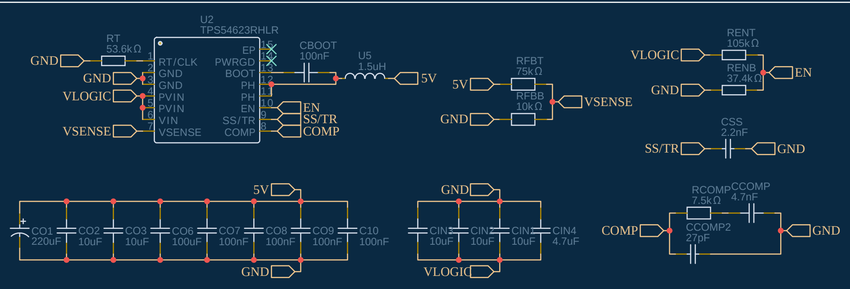
Battery Balancing IC



Battery Charger

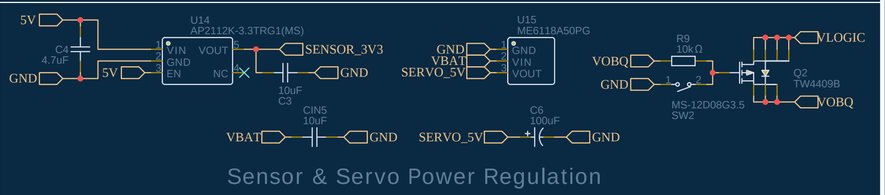


Pi5 Power Regulator Circuit



*Creates a 2S configuration from 2 separate batteries that is fused (4A sustained, 8A trip, resettable). Fed to the main voltage regulator (6-9V input, ~5V output), and later to the decoupling network that is made up of various LF/HF filtering ceramic caps and bulk tantalum caps to ensure stability of the 5V rail at all times.

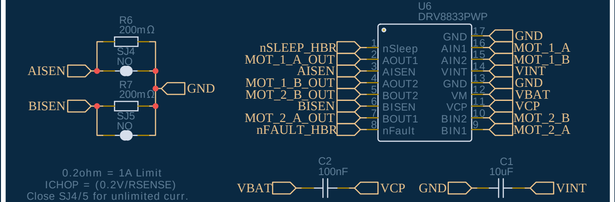
Sensor & Servo Power Regulation



Power Regulation and Control

H-BRIDGE Circuit

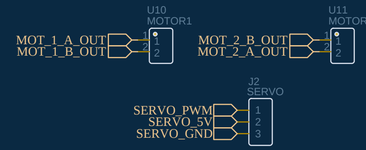
DRV8833 with 1A limit



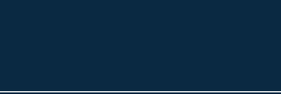
0.2ohm = 1A Limit
ICHOP = (0.2V/RSENSE)
Close SJ4/5 for unlimited curr.

Motor Interface

N30 Movement Motor + Steering Servo



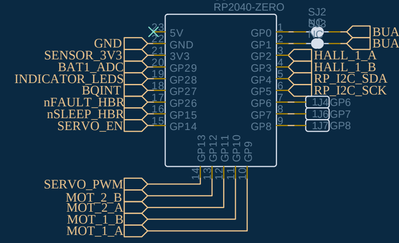
Motor Enable Logic



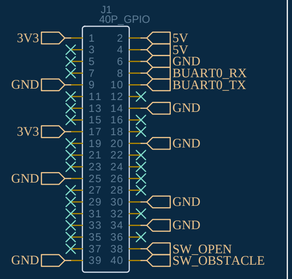
Motor Control

Raspberry Pi Pico Tiny

RP2040/2350 Minimalistic MCU



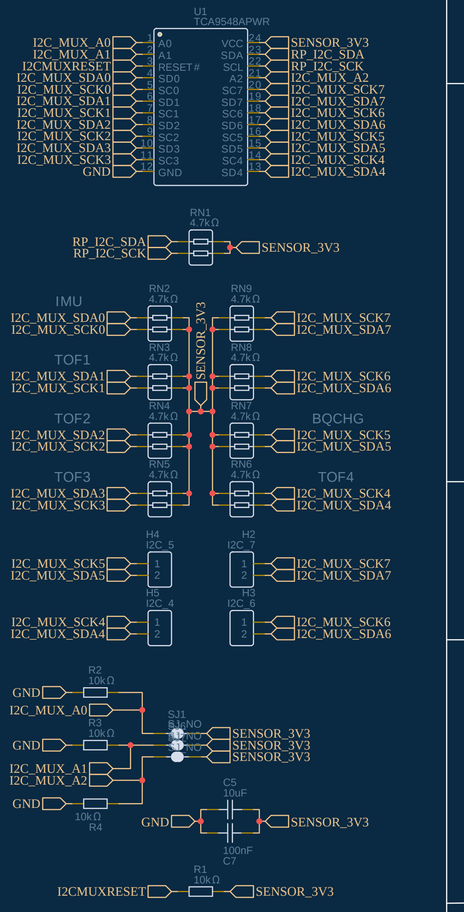
Pi5 GPIO Interface



Robot Brain

RP2350-Zero & Rpi 5 8GB

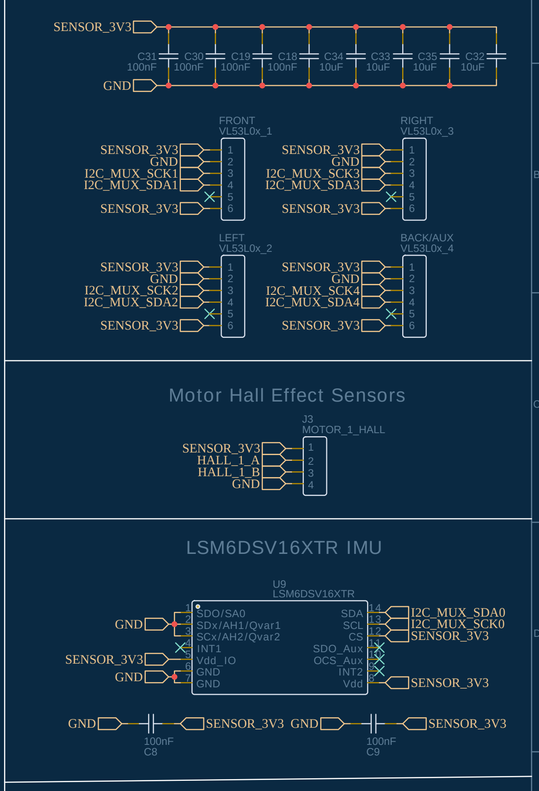
I2C Multiplexer





Battery Voltage Sensing

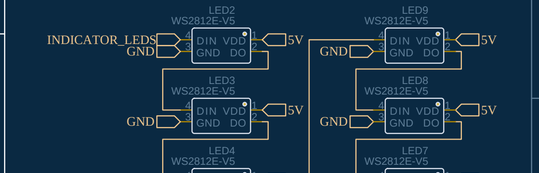
VL53L1 ToF



Motor Hall Effect Sensors



LSM6DSV16XTR IMU



LED Indicators

Battery charge monitor + status monitor

TITLE: WRO2026 FE Robot for Euro		REV: 3.0
EasyEDA	Company: SETŠ, Team Elektron	Sheet: 1/1
	Date: 2026-01-03	Drawn By: Team Elektron